

The mathematical bridge connecting the observed, unlinked marginals (Panel A) and the recovered latent SGPC regime (Panel C) has two ingredients:

1. A baseline copula  $C_0$  providing a dependence template from prior reference-quantiles  $U$  to current reference-quantiles  $V$ .<sup>2</sup>
2. A stochastic growth regime  $G$  that governs how subgroup  $S$  allocates conditional percentiles within that template.<sup>3, 4</sup>

Together,  $C_0$  and  $G$  define a *forward map* from a candidate regime to a *predicted* current-grade marginal CDF. Panel B2 visualizes this map via the induced  $F_G(v)$ , and Panel B1 *inverts* it by choosing the regime whose  $F_G$  best matches the observed current marginal  $F_{V,S}^{\text{obs}}$ .

Let  $C_0$  be a baseline copula.  $C_0$  induces the conditional CDF (transition kernel)

$$F_0(v | u) = \Pr(V \leq v | U = u) = \frac{\partial}{\partial u} C_0(u, v).$$

Its inverse in  $v$  (the conditional quantile kernel) is

$$Q_0(p | u) = F_0^{-1}(p | u).$$

Intuitively, for each starting quantile  $u$ , the kernel  $F_0(\cdot | u)$  describes the distribution of possible ending quantiles  $V$  under the baseline dependence structure  $C_0$ .

Let  $P \in [0, 1]$  be a random variable (a latent SGPC on the 0–1 scale) and let  $p$  denote a realization in  $[0, 1]$ . A *growth regime*  $G$  is the distribution of  $P$ , represented by its CDF

$$G(p) := \Pr(P \leq p), \quad p \in [0, 1].$$

Given  $U = u$  and a draw  $P = p$ , the model generates an outcome percentile by:<sup>5</sup>

$$V = Q_0(P | U = u).$$

The uniform regime  $G(p) = p$  (i.e.,  $P \sim \text{Uniform}(0, 1)$ ) corresponds to occupying the baseline kernel with no tilt. Departures from  $G(p) = p$  tilt the growth regime mass toward lower or higher conditional percentiles relative to the baseline regime.

Although we do not observe linked pairs  $(u, v)$  in TIMSS- or NAEP-style settings, the model implies a forward map from the observed prior marginal  $F_U$  to the predicted current marginal  $F_G$ . For any fixed  $u$ ,

$$\begin{aligned} \Pr(V \leq v | U = u) &= \Pr(Q_0(P | u) \leq v) \\ &= \Pr(P \leq F_0(v | u)) \\ &= G(F_0(v | u)). \end{aligned}$$

Therefore, by the law of total probability, the subgroup's predicted current-grade marginal CDF is

$$\begin{aligned} F_G(v) &= \Pr(V \leq v) \\ &= \mathbb{E}_U [G(F_0(v | U))] \\ &= \int_0^1 G(F_0(v | u)) dF_U(u) \\ &\approx \frac{\sum_{i=1}^n w_i G(F_0(v | u_i))}{\sum_{i=1}^n w_i}, \end{aligned}$$

where  $\{u_i, w_i\}$  are the observed prior pseudo-observations and survey weights for the subgroup (i.e., a weighted empirical approximation to  $F_U$ ).

We estimate the subgroup's growth regime by searching over an admissible class  $\mathcal{G}$  of candidate CDFs on  $[0, 1]$ . Panel B1 shows the heat map of  $\mathcal{G}$  over the *Beta regime family*

$$\mathcal{G}_{\text{Beta}} = \{G_{m,\kappa}(\cdot) : P \sim \text{Beta}(\kappa m, \kappa(1 - m)), 0 < m < 1, \kappa > 0\},$$

where  $m = \mathbb{E}[P]$  controls the regime center (mean SGPC on the 0–1 scale) and  $\kappa$  controls concentration (dispersion of latent SGPC).<sup>6</sup>

For each candidate  $G \in \mathcal{G}$ , Panel B2 provides the induced *predicted* current-grade marginal

$$F_G(v) = \mathbb{E}_U [G(F_0(v | U))].$$

The growth regime estimate for subgroup  $S$  is defined by the minimum-distance problem

$$\hat{G}_S = \underset{G \in \mathcal{G}_{\text{Beta}}}{\text{argmin}} W_1(F_{V,S}^{\text{obs}}, F_G),$$

where

$$W_1(F_1, F_2) = \int_0^1 |F_1(v) - F_2(v)| dv$$

is the Wasserstein-1 (earth mover's) distance between the observed and predicted current-grade marginal CDFs.<sup>7</sup>

The heat map displays  $\log_{10}$  Wasserstein-1 distance over  $(m, \log_{10} \kappa)$ : lighter regions indicate candidate regimes whose induced  $F_G$  more closely matches  $F_{V,S}^{\text{obs}}$ . The marked optimum corresponds to  $\hat{G}_S$  (equivalently,  $(\hat{m}, \hat{\kappa})$ ). For reference, the uniform regime  $G(p) = p$  occurs at  $m = 0.5$  and  $\kappa = 2$  (i.e.,  $\text{Beta}(1, 1)$ ), indicated by the  $U(0, 1)$  marker.

The baseline copula,  $C_0$ , combined with the best fit growth regime estimate,  $\hat{G}_S$ , induces the Wasserstein-1 minimized CDF,  $F_{\hat{G}_S}(v)$ , and its density,  $f_{\hat{G}_S}(v)$ , which are shown in panel B2 and A, respectively, with a dashed-black line.